CHEMICAL CHANGE

Physical Science

Quarter 3

Objective:

- Use simple collision theory to explain the effects of concentration, temperature, and particle size on the rate of reaction (S11/12PS-IIIf-23)

Chemical Equation

- In chemistry, a chemical equation is needed to find the amount of materials (**reactants**) that you need and the new substance (**product**) to be formed.

$$\text{Reactants} \xrightarrow{\text{React to form}} \text{Products}$$

Collision Theory of Chemical Reaction

- the greater the frequency of collisions, the higher the rate of reaction

Collisions resulting in a reaction to form a product is called **effective collision**.

- Do all collisions result in the formation of products?

FACTORS AFFECTING REACTION RATES

- Reactant Concentration
- Increase in Temperature
- Increase in Surface Area

Reactant Concentration

- **At a higher concentration, the molecules collide more often.**
- This results in an increase in the number of effective collisions thereby resulting in increase reaction rates.

Let's Connect!

- Give sample applications that show how concentration affect the rate of chemical reaction.

Increase in Temperature

- **When temperature becomes higher, the molecules of the reactants move faster.**
- This causes more frequent collision. It also results in greater energy at impact.
- These factors increase the number of effective collisions and an increase in the rate of reaction.

Let’s Connect!

- Give sample applications that show how temperature affect the rate of chemical reaction.

Increase in the Surface Area at Solid Reactants

- **With greater surface area, the molecules of the reactants can collide more frequently**, thereby resulting in greater number of effective collisions and increase in reaction rates.

Example: Detergent bar vs. Detergent powder

Let’s Connect!

- Give sample applications that show how particle size affect the rate of chemical reaction.

Properly Oriented Molecules

- Molecules must be oriented such that it favors reaction.

Example: If the reactants are Cl and NOCl, collision between the two molecules will more effective when the Cl atom collides with the Cl atom of NOCl molecule.

The formation of the products Cl2 and NO will occur only if the collision is effective. Otherwise, the reactants will only bounce off with one another.

Group Activity 1:

Explain the effects of the following on the rate of chemical reaction:

1. Concentration
2. Temperature
3. Particle size

Directions:

- Form groups based on your sweepers grouping.
- Each group will present an illustration on how each factor affects the rate of chemical reaction and at least 1 example.

Assignment (1/2 sheet of paper) Study and answer:

- What are the factors affect the rate of chemical reaction?
- Explain how each factor affects the rate of chemical reaction.